

KEONJHAR DISTRICT

Household Survey 2026

Data Cleaning | Power BI Dashboard Documentation

Project	Keonjhar Household Survey – Data Cleaning & Dashboard
Version	v2.0 – Final
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Data Source	Keonjhar Household Survey (Odia)
Raw File	Keonjhar_household_survey_Odia_all_versions_labels_2026-05-04.xlsb
Total Records	67,307 Households 2,38,471 Individuals
Tool Used	Power BI Desktop

1. Project Overview

This document covers the complete workflow for the Keonjhar District Household Survey 2026 — from raw KoBoToolbox export through Python data cleaning to final Power BI dashboard development. The survey covers 67,307 households and 2,38,471 individuals across 14 blocks, 174 Gram Panchayats and 662 villages of Keonjhar district, Odisha.

1.1 Why Python First?

The raw survey file is a 42 MB .xlsb (Excel Binary Workbook). Power BI cannot load this file efficiently — it takes 10+ minutes because .xlsb uses a compressed binary format that Power BI must parse cell by cell. Python reads the file once, cleans it, and exports clean CSV files that Power BI loads in under 5 seconds.

1.2 Dashboard Structure

The Power BI report has 5 main dashboard pages accessible from a clickable navigation home page:

Page	Title	Core Question It Answers
Home	Navigation Page	Entry point — click any section to jump to that report
1	Overview & Geography	Where were households surveyed? How many per block/village?
2	Demographics & Housing	Who are these people? What are their living conditions?
3	Health & Nutrition	Are they protected from illness? Are children malnourished?
4	Education & Livelihoods	Are children learning? How do households earn and survive?
5	Women & Social Protection	Are women and vulnerable groups included in schemes?

2. Python Data Cleaning

All cleaning was done in Python using the pandas and pyxlsb libraries. The script runs in one step from raw .xlsb to clean .csv output.

2.1 Install Required Libraries

```
pip install pandas pyxlsb openpyxl
```

2.2 Step-by-Step Cleaning Code

STEP 1 — Load the file

```
import pandas as pd, pyxlsb, re, warnings
from datetime import datetime, timedelta
warnings.filterwarnings('ignore')

# dtype=object prevents pandas from dropping date/numeric values
df_main = pd.read_excel('Keonjhar_survey.xlsb',
                        engine='pyxlsb',
                        sheet_name='Keonjhar household survey ( ...',
                        dtype=object)
```

```
df_mem = pd.read_excel('Keonjhar_survey.xlsx',  
    engine='pyxlsb',  
    sheet_name='member_details',  
    dtype=object)  
  
print(f'Main: {df_main.shape}    Member: {df_mem.shape}')
```

STEP 2 — Remove completely empty columns

Columns with zero data in every row have no value in any dashboard. KoBoToolbox always adds warning/display fields (_notes, _tags, show_block_code, etc.) that are completely blank.

```
df_main.dropna(axis=1, how='all', inplace=True)  
df_mem.dropna(axis=1, how='all', inplace=True)
```

STEP 3 — Remove KoBoToolbox system columns

These are internal system columns added by KoBoToolbox — not survey answers. The 'duration' column is 99.7% null. The 'gps' column is a compound string that duplicates the four separate GPS columns already in the data.

```
drop_cols = ['_id', '_submission_time', 'gps', 'duration',  
    '_status', '__version__', 'meta/rootUuid',  
    '_submission_id', '_parent_table_name']  
  
for col in drop_cols:  
    if col in df_main.columns: df_main.drop(columns=[col], inplace=True)  
    if col in df_mem.columns: df_mem.drop(columns=[col], inplace=True)
```

STEP 4 — Remove multi-select sub-columns

KoBoToolbox creates one column per checkbox for multi-select questions, named like 'Reason_for_dropout/Financial burden'. These are 0/1 binary flags. The parent column already contains the full answer as a readable string. Sub-columns contain '/' in their name.

```
df_main.drop(columns=[c for c in df_main.columns if '/' in str(c)], inplace=True)  
df_mem.drop(columns=[c for c in df_mem.columns if '/' in str(c)], inplace=True)
```

STEP 5 — Fix date columns (Excel serial → proper datetime)

The start and end columns are stored as Excel floating-point serial numbers (e.g. 46113.259). Power BI cannot use these as dates. We convert them and also create two helper columns: interview_date (date only for slicers) and interview_duration_min (how long the survey took).

```
def convert_excel_date(number):  
    try:  
        return datetime(1899, 12, 30) + timedelta(days=float(number))  
    except:  
        return None  
  
df_main['start'] = df_main['start'].apply(convert_excel_date)  
df_main['start'] = pd.to_datetime(df_main['start']).dt.strftime('%Y-%m-%d %H:%M:%S')  
  
df_main['end'] = df_main['end'].apply(convert_excel_date)  
df_main['end'] = pd.to_datetime(df_main['end']).dt.strftime('%Y-%m-%d %H:%M:%S')
```

```
df_main['interview_date'] = pd.to_datetime(df_main['start']).dt.strftime('%Y-%m-%d')
df_main['interview_duration_min'] = (
    (pd.to_datetime(df_main['end']) - pd.to_datetime(df_main['start']))
    .dt.total_seconds() / 60).round(1)
```

STEP 6 — Strip whitespace from text columns

Many cells have trailing spaces e.g. 'Jhumpura ' instead of 'Jhumpura'. In Power BI these appear as two separate values in every filter and chart, breaking all group-by visuals. The try/except skips date and numeric columns that do not support .str operations.

```
for col in df_main.columns:
    if df_main[col].dtype == object:
        try:
            df_main[col] = df_main[col].str.strip()
        except:
            pass

for col in df_mem.columns:
    if df_mem[col].dtype == object:
        try:
            df_mem[col] = df_mem[col].str.strip()
        except:
            pass
```

STEP 7 — Standardise column names

All column names are converted to lowercase `_snake_case`. Spaces and special characters break DAX formulas in Power BI. The `household_index` column is the JOIN key between the two tables.

```
def clean_name(name):
    name = str(name).strip().lower()
    name = re.sub(r'[\s\~]+', '_', name)
    name = re.sub(r'[^\\w]', '', name)
    name = re.sub(r'_+', '_', name)
    return name.strip('_')

df_main.columns = [clean_name(c) for c in df_main.columns]
df_mem.columns = [clean_name(c) for c in df_mem.columns]

# Rename the JOIN key column - same name in both tables
df_main.rename(columns={'_index': 'household_index'}, inplace=True)
df_mem.rename(columns={'_parent_index': 'household_index'}, inplace=True)
```

STEP 8 — Remove duplicate rows

```
df_main.drop_duplicates(inplace=True)
df_mem.drop_duplicates(inplace=True)
```

STEP 9 — Export to CSV for Power BI

CSV files load in seconds in Power BI. utf-8-sig encoding ensures Odia and Hindi text displays correctly. Load both files in Power BI and connect them via `household_index`.

```
df_main.to_csv('main_survey.csv', index=False, encoding='utf-8-sig')
df_mem.to_csv('member_details.csv', index=False, encoding='utf-8-sig')
```

```
print(f'main_survey.csv    -> {df_main.shape[0]:,} rows x {df_main.shape[1]} cols')
print(f'member_details.csv -> {df_mem.shape[0]:,} rows x {df_mem.shape[1]} cols')
print('Done! Load both CSVs in Power BI.')
```

2.3 Cleaning Summary

Metric	Before Cleaning	After Cleaning
Main sheet rows	67,307	67,307 (no rows lost)
Main sheet columns	202	~139 columns
Member sheet rows	2,38,471	2,38,471 (no rows lost)
Member sheet columns	73	~36 columns
Empty columns removed	—	15 main + 9 member
Multi-select sub-cols	—	43 main + 22 member removed
Whitespace cells fixed	—	52,818+ cells
Duplicate rows	0	0
Date columns	Float numbers	Proper datetime strings
File format	42 MB .xlsb	~60 MB CSV (loads in 5 sec)

3. Power BI Setup

3.1 Load Data

- Open Power BI Desktop
- Home → Get Data → Text/CSV → select main_survey.csv → Load
- Home → Get Data → Text/CSV → select member_details.csv → Load

3.2 Create Table Relationship

- Go to Model view (left panel)
- Drag household_index from main_survey onto household_index in member_details
- Relationship: Many (member_details) to One (main_survey)
- Cross filter direction: Single

3.3 Common Slicers on Every Page

Add these 6 slicers at the top of every page so users can filter by location and category across all visuals:

Slicer	Column to Use
Block	main_survey[block]
District	main_survey[district]
Gram Panchayat	main_survey[gram_panchayat]
Village	main_survey[village_name]
Interview Date	main_survey[interview_date]
Social Category	main_survey[social_category]

Tip: Right click each slicer → Format → Style → Dropdown. This saves space on every page.

4. Dashboard Pages — Visuals, Insights & DAX

Page 1 — Overview & Geography

Core Question: Where were households surveyed? What is the geographic reach of the survey?

What this page shows:

- Total households, individuals, blocks, villages and GPs covered — shown as KPI cards
- Households per block — horizontal bar chart to compare survey coverage
- Population per block and village — treemap showing relative size
- Detailed block-wise breakdown table: total population, male, female, ST/SC/OBC, youth, senior citizens, children
- Slicers: Block, District, Gram Panchayat

How this helps:

- Government and NGO planners can instantly see which blocks have highest/lowest survey coverage
- Identifying underrepresented villages for follow-up surveys
- Comparing social category distribution across blocks helps target welfare schemes geographically

DAX Measures — Page 1:

Measure Name	DAX Formula	Purpose
Total Households	COUNTROWS (main_survey)	Count of all surveyed households
Total Population	COUNTROWS (member_details)	Count of all individuals
Total Blocks	DISTINCTCOUNT (main_survey[block])	Unique blocks covered
Total Villages	DISTINCTCOUNT (main_survey[village_name])	Unique villages covered
Total GPs	DISTINCTCOUNT (main_survey[gram_panchayat])	Unique Gram Panchayats
Avg Household Size	AVERAGE (main_survey[total_household_member])	Average members per household
ST Households	CALCULATE (COUNTROWS (main_survey), main_survey[social_category]="ST")	Households from ST category
SC Households	CALCULATE (COUNTROWS (main_survey), main_survey[social_category]="SC")	Households from SC category
Youth Count	CALCULATE (COUNTROWS (member_details), member_details[member_age]>=15, member_details[member_age]<=35)	Members aged 15 to 35
Senior Citizens	CALCULATE (COUNTROWS (member_details), member_details[member_age]>=60)	Members aged 60 and above

Page 2 — Demographics & Housing

Core Question: Who are these people and how do they live?

What this page shows:

- % ST population, % female, % disabled — headline KPI cards
- PMAY coverage %, % Kutcha houses, % with toilet — housing KPI cards
- House type per block (Kutcha / Pucca / Semi-Pucca) — stacked bar chart
- Total household breakdown by social category — donut chart
- Disability profile by social category — donut chart
- Toilet, LPG and electricity coverage % per block — grouped bar chart
- Slicers: Block, Gender, Gram Panchayat

How this helps:

- Identifies blocks with high proportion of Kutcha houses for housing scheme targeting
- Toilet and electricity coverage gaps visible at block level for infrastructure planning
- Disability data by social category helps target disability certificates and pensions
- PMAY coverage gaps show where housing scheme has not reached eligible households

DAX Measures — Page 2:

Measure Name	DAX Formula	Purpose
ST %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[social_category]="ST"),COUNTROWS (main_survey))</code>	ST household as % of total
Female %	<code>DIVIDE (CALCULATE (COUNTROWS (member_details),member_details[member_gender]="Female"),COUNTROWS (member_details))</code>	Female members as %
Disabled %	<code>DIVIDE (CALCULATE (COUNTROWS (member_details),member_details[member_disability_present]="Yes"),COUNTROWS (member_details))</code>	Members with disability
PMAY %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[pmay_benefit]="Yes"),COUNTROWS (main_survey))</code>	PMAY beneficiary households
Kutcha Houses %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[house_type]="Kuccha"),COUNTROWS (main_survey))</code>	% households in Kutcha houses
Toilet %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[toilet_available]="Yes"),COUNTROWS (main_survey))</code>	Households with toilet
LPG %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[lpg_connection]="Yes"),COUNTROWS (main_survey))</code>	Households with LPG

Electricity %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[electricity_metered]="Yes"),COUNTROWS (main_survey))</code>	Households with metered electricity
Piped Water %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[pipe_water_supply]="Yes"),COUNTROWS (main_survey))</code>	Households with piped water

Page 3 — Health & Nutrition

Core Question: Are people protected from illness? Are children malnourished?

What this page shows:

- Uninsured household members count, SAM children, MAM children — KPI cards
- % immunised children, % NCD screened, % institutional delivery — KPI cards
- Chronic disease members by block — horizontal bar chart
- Drinking water source by population — bar chart
- Food consumption frequency for 6 food groups (meat, rice, eggs, vegetables, millet, fruits) — small multiples bar charts
- Slicers: Block, District, Gram Panchayat

How this helps:

- SAM and MAM counts identify blocks needing urgent nutrition intervention
- Low immunisation % and NCD screening % highlight health access gaps
- Drinking water source data identifies communities relying on unsafe water
- Food frequency charts reveal dietary diversity gaps for nutrition programmes
- Uninsured member count helps target BSKY/PMJAY enrollment drives

DAX Measures — Page 3:

Measure Name	DAX Formula	Purpose
Uninsured HH Members	<code>SUM(main_survey[uninsured_household_members])</code>	Total uninsured individuals
SAM Children	<code>SUM(main_survey[sam_children_count])</code>	Severely malnourished children
MAM Children	<code>SUM(main_survey[mam_children_count])</code>	Moderately malnourished children
Immunised %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[child_immunisation]="Yes"),COUNTROWS (main_survey))</code>	Child immunisation coverage

NCD Screened %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[ncd_screening_done]="Yes"),COUNTROWS (main_survey))</code>	NCD screening coverage
Inst. Delivery %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[institutional_delivery]="Yes"),COUNTROWS (main_survey))</code>	Institutional delivery rate
Malnutrition Rate %	<code>DIVIDE (SUM (main_survey[sam_children_count])+SUM (main_survey[mam_children_count]),SUM (main_survey[total_under5_children]))</code>	SAM+MAM out of under-5
Insurance %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[household_health_insurance]="Yes"),COUNTROWS (main_survey))</code>	Insured households
Chronic Disease %	<code>DIVIDE (CALCULATE (COUNTROWS (member_details),member_details[member_chronic_disease]<>"None"),COUNTROWS (member_details))</code>	Members with chronic disease
Cancer Screening %	<code>DIVIDE (CALCULATE (COUNTROWS (main_survey),main_survey[cancer_screening_done]="Yes"),COUNTROWS (main_survey))</code>	Cancer screening coverage

Page 4 — Education & Livelihoods

Core Question: Are children learning? How do households earn and survive?

What this page shows:

- Children in school, total dropouts, dropout %, literacy rate % — KPI cards
- Scholarship children count, training to job conversion % — KPI cards
- Migrant % and agriculture % — additional KPI cards
- Dropout reasons — horizontal bar chart showing all reasons ranked by frequency
- Gender breakdown for: in school, school dropouts, literacy, employed after training — donut charts
- Slicers: Block, Gender, Education Level

How this helps:

- Financial burden as top dropout reason helps design targeted scholarship interventions
- Gender disparity in literacy and dropouts identifies blocks needing girl-child education support
- Training to job conversion rate shows effectiveness of skill training programmes
- Agriculture % and migrant % reveal livelihood patterns for employment scheme planning
- Scholarship data shows reach of government education welfare schemes

DAX Measures — Page 4:

Measure Name	DAX Formula	Purpose
Children in School	<code>SUM (main_survey[children_in_school])</code>	Total children currently enrolled

Total Dropouts	SUM(main_survey[school_dropout_children])	Total school dropout children
Dropout %	DIVIDE(SUM(main_survey[school_dropout_children]),SUM(main_survey[children_in_school])+SUM(main_survey[school_dropout_children]))*100	Dropout rate
Literacy Rate %	DIVIDE(CALCULATE(COUNTROWS(member_details),member_details[member_education_level]<>"Illiterate",member_details[member_age]>6),CALCULATE(COUNTROWS(member_details),member_details[member_age]>6))	Literate members above age 6
Scholarship Children	SUM(main_survey[scholarship_children_count])	Children receiving scholarship
Training to Job %	DIVIDE(CALCULATE(COUNTROWS(main_survey),main_survey[employment_after_training]="Yes"),CALCULATE(COUNTROWS(main_survey),main_survey[member_training_received]="Yes"))	Training employment conversion
Migrant %	DIVIDE(CALCULATE(COUNTROWS(member_details),member_details[member_migrant_worker]="Yes"),COUNTROWS(member_details))	Migrant workers as % of total
Agriculture %	DIVIDE(CALCULATE(COUNTROWS(main_survey),main_survey[agriculture_practiced]="Yes"),COUNTROWS(main_survey))	Households practicing agriculture
MGNREGA %	DIVIDE(CALCULATE(COUNTROWS(member_details),member_details[member_mgnrega_jobcard]="Yes"),COUNTROWS(member_details))	Members with MGNREGA job card
Avg Annual Income	AVERAGE(main_survey[household_annual_income])	Average household annual income

Page 5 — Women & Social Protection

Core Question: Are women and vulnerable groups included in welfare schemes?

What this page shows:

- Total women, SHG members %, MAM children count — KPI cards
- % immunised children, % NCD screened, % institutional delivery — KPI cards
- Subhadra beneficiary vs Subhadra eligible — comparison bar chart showing the gap
- SHG members vs SHG revolving fund received by block — grouped bar chart
- Smartphone users, digital payment users and Aadhaar bank linked by block — grouped bar chart
- Slicers: Block, District, Gram Panchayat

How this helps:

- Subhadra gap (eligible but not yet beneficiary) identifies women to be enrolled urgently
- SHG revolving fund gap shows blocks where SHG groups formed but funds not yet received
- Digital payment and smartphone data informs digital literacy programme targeting
- Aadhaar bank linkage gaps identify women who may miss direct benefit transfers
- Block-level SHG data helps monitor women's self-help group programme implementation

DAX Measures — Page 5:

Measure Name	DAX Formula	Purpose
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Total Women	CALCULATE (COUNTROWS (member_details),member_details[member_gender]="Female")	Total female members
SHG Members %	DIVIDE (CALCULATE (COUNTROWS (member_details),member_details[member_shg_membership]="Yes"),CALCULATE (COUNTROWS (member_details),member_details[member_gender]="Female")) *100	SHG membership among women
Subhadra Eligible	SUM(main_survey[subhadra_eligible_women_count])	Women eligible for Subhadra Yojana
Subhadra Beneficiary	SUM(main_survey[subhadra_beneficiary_women_count])	Women receiving Subhadra benefit
Subhadra Coverage %	DIVIDE (SUM(main_survey[subhadra_beneficiary_women_count]),SUM(main_survey[subhadra_eligible_women_count])) *100	Subhadra scheme coverage
Subhadra Gap	SUM(main_survey[subhadra_eligible_women_count]) - SUM(main_survey[subhadra_beneficiary_women_count])	Eligible but not yet beneficiary
SHG Revolving Fund	CALCULATE (COUNTROWS (main_survey),main_survey[shg_revolving_fund_received]="Yes")	SHGs that received revolving fund
Aadhaar Bank %	DIVIDE (CALCULATE (COUNTROWS (member_details),member_details[member_aadhaar_bank_linked]="Yes"),COUNTROWS (member_details))	Members with Aadhaar-bank link
Digital Payment %	DIVIDE (CALCULATE (COUNTROWS (member_details),member_details[member_digital_payment_user]="Yes"),COUNTROWS (member_details))	Digital payment users %

Smart phone %	DIVIDE (CALCULATE (COUNTROWS (member_details),member_details[member_smartphone_user]="Yes"),COUNTROWS (member_details))	Smart phone users %
Wome n Pensi on Gap	CALCULATE (COUNTROWS (member_details),member_details[member_pension_eligible]="Yes",member_details[member_pension_receiving]="No",member_details[member_gender]="Female")	Wom en eligibl e but not receiv ing pensi on